

Computing Platform for Autonomous Driving (CP4AD)

L1. RTOS #1

Prof. Jong-Chan Kim

Dept. Automobile and IT Convergence



국민대학교
KOOKMIN UNIVERSITY

Erika Enterprise



evidence / erika3 Public

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eguidieri [tricore] Fixed interrupt regression in... 85c445d on 13 Sep 2019 245 commits

contrib	[kernel] PostTaskHook called twice fixed and Initial ...	3 years ago
doc	[kernel] Fixed WaitEvent, in case of multicore enviro...	4 years ago
mk	[kernel] Fixed WaitEvent, in case of multicore enviro...	4 years ago
pkg	[tricore] Fixed interrupt regression introduced by co...	3 years ago
LICENSE.TXT	Update LICENSE to all files	4 years ago
README.md	[kernel] Fixed WaitEvent, in case of multicore enviro...	4 years ago
THIRDPARTY.TXT	[THIRDPARTY] Rephrased and fixed the statement r...	3 years ago

About

ERIKA Enterprise v3 RTOS

www.erika-enterprise.com

arduino avr arm real-time
cortex-m x86-64 xen rtos
cortex-a autosar osek dspic
aurix tricore-development cortex-r5f
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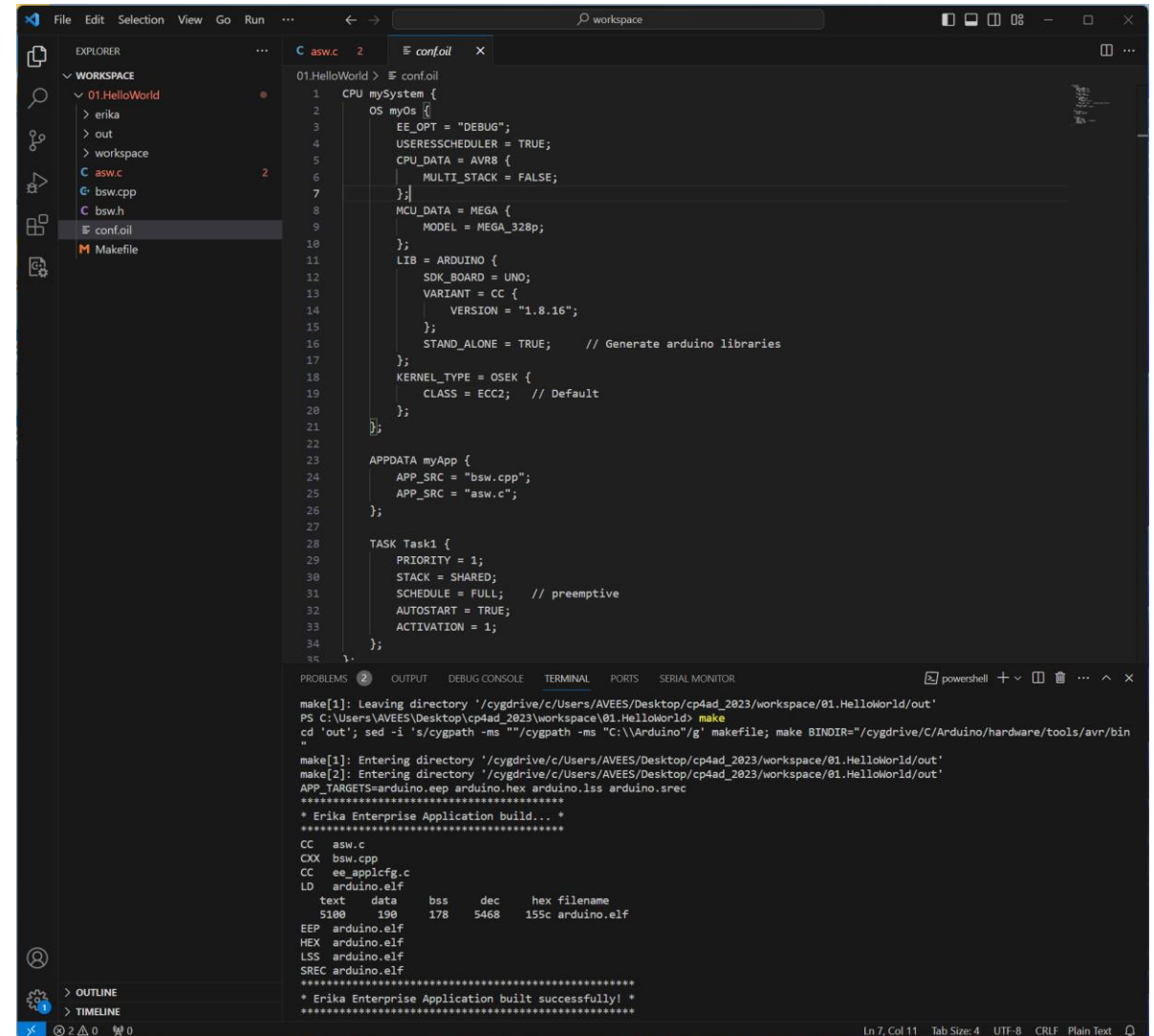
[ERIKA WIKI](#)



- 이탈리아 EVIDENCE에서 개발된 오픈소스 OSEK/VDX RTOS 구현체
- 듀얼 라이선스 정책 (오픈소스 라이선스 + 상용 라이선스)
- RTOS 연구 및 교육에 널리 사용
- GitHub을 통한 코드 배포 or RTDruide와 통합된 이클립스 설치

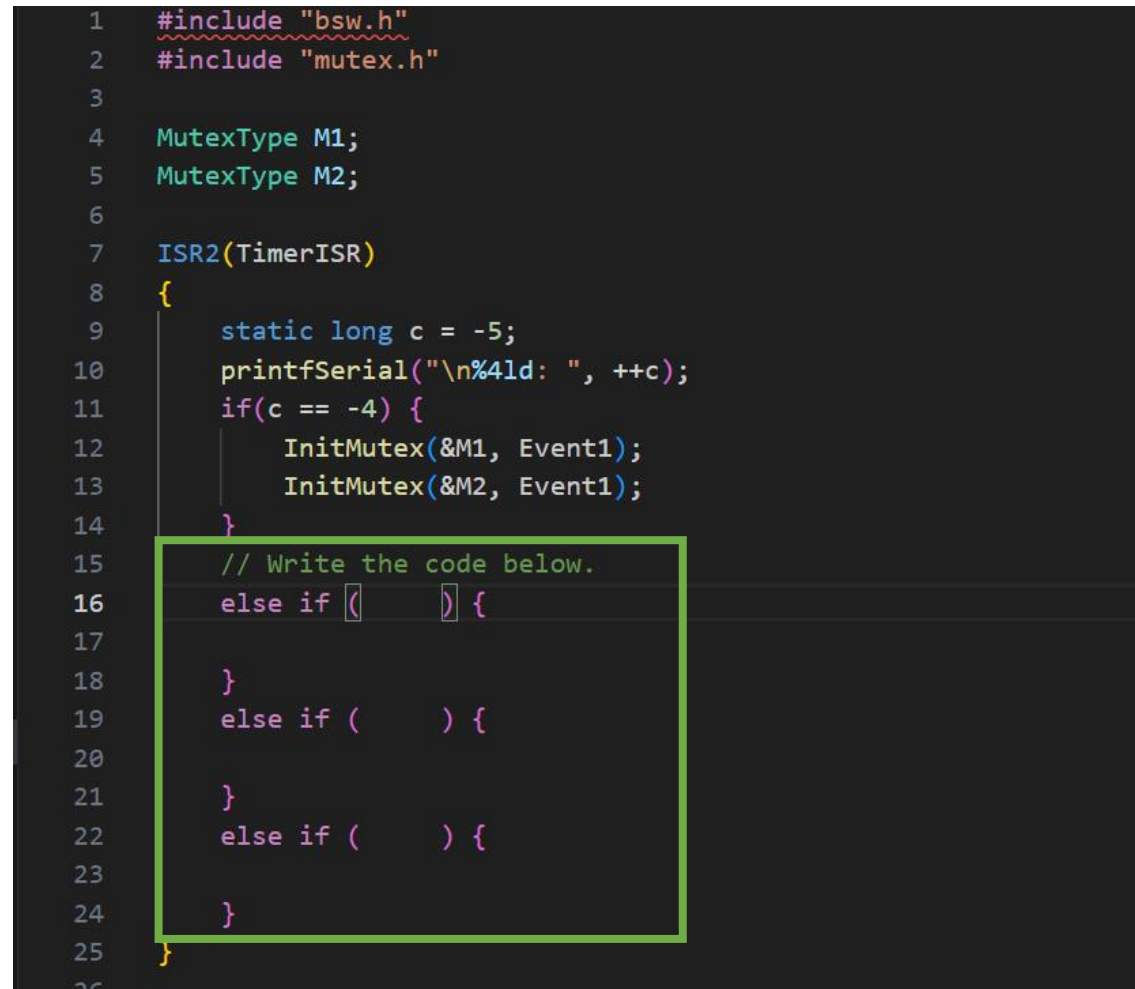
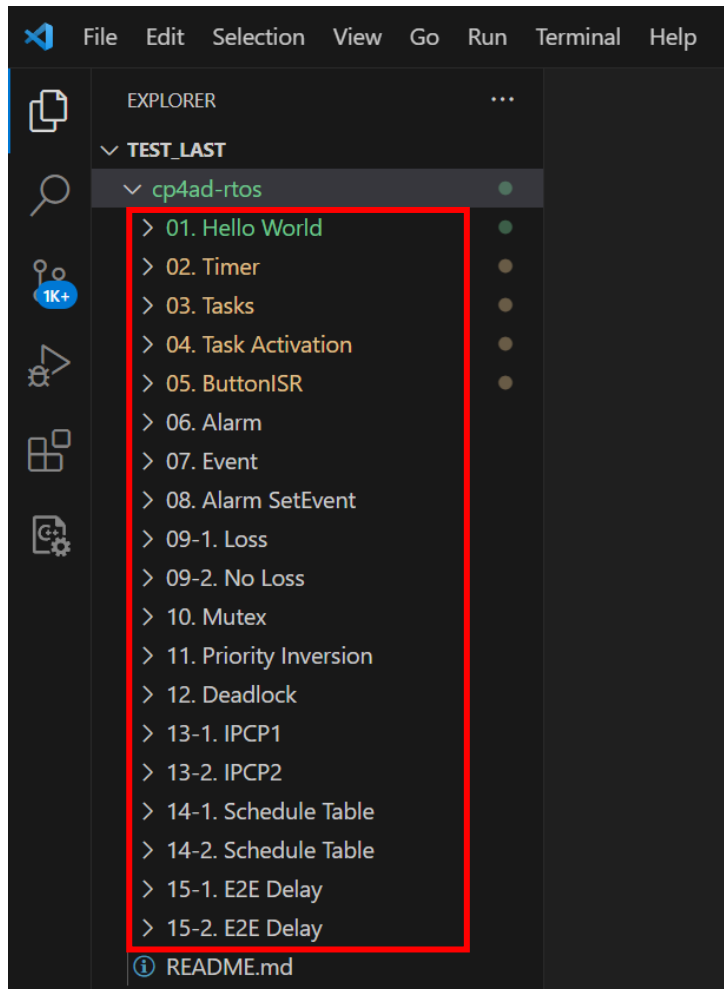
Visual Studio Code

- 프로젝트 생성 및 관리
- OIL 파일, C/C++ 파일 편집
- 프로젝트 빌드
- 실행파일 업로드
- 터미널 실행



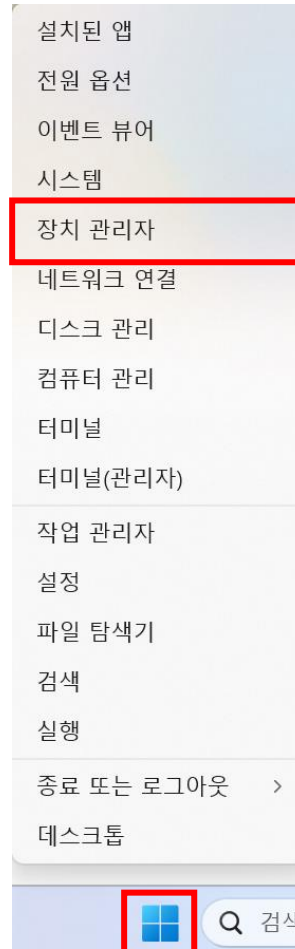
CP4AD RTOS Practice github

- <https://github.com/AveesLab/cp4ad-rtos.git> → workspace에 폴더 생성
- **// Write the code below.** 아래 코드 추가하는 방식으로 실습 진행

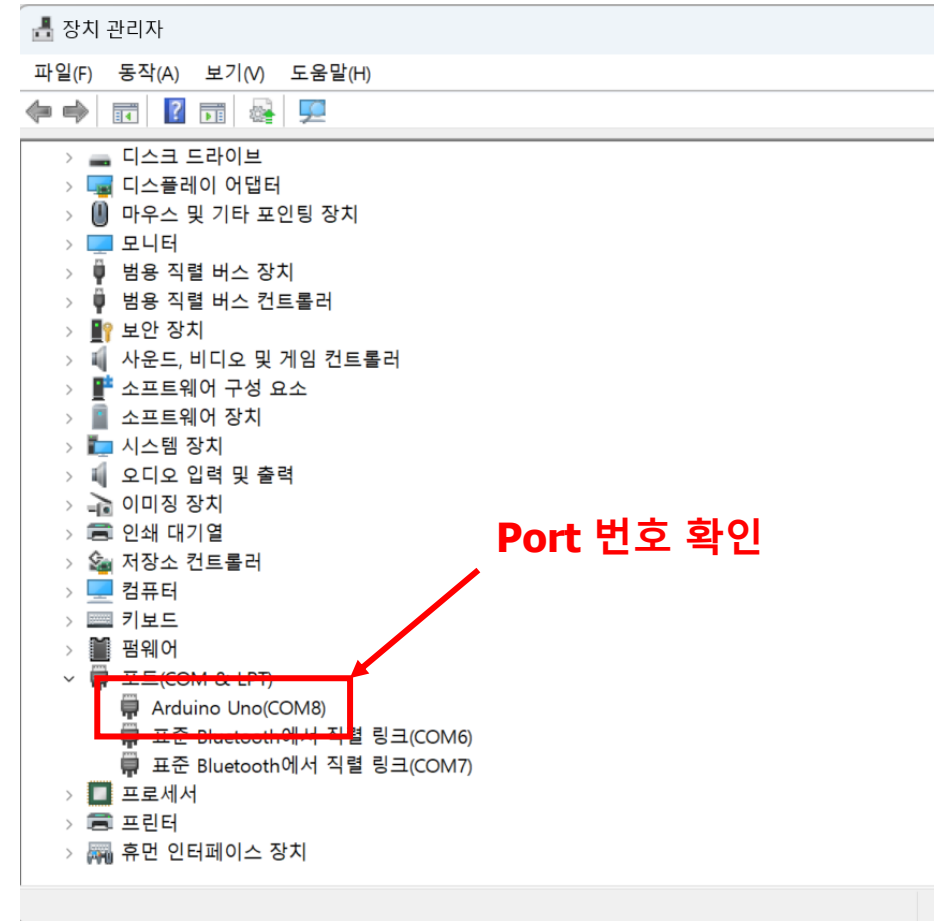


Erika Enterprise Build Process

- Arduino 연결 후 포트 번호 확인(장치 관리자 > 포트)



우클릭



Erika Enterprise Build Process

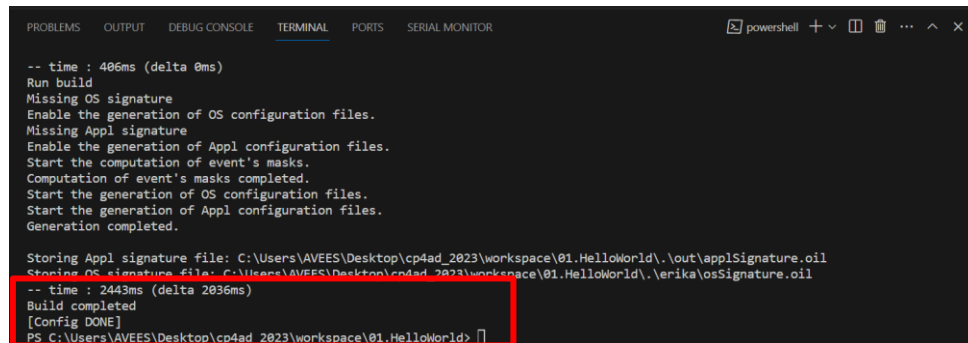
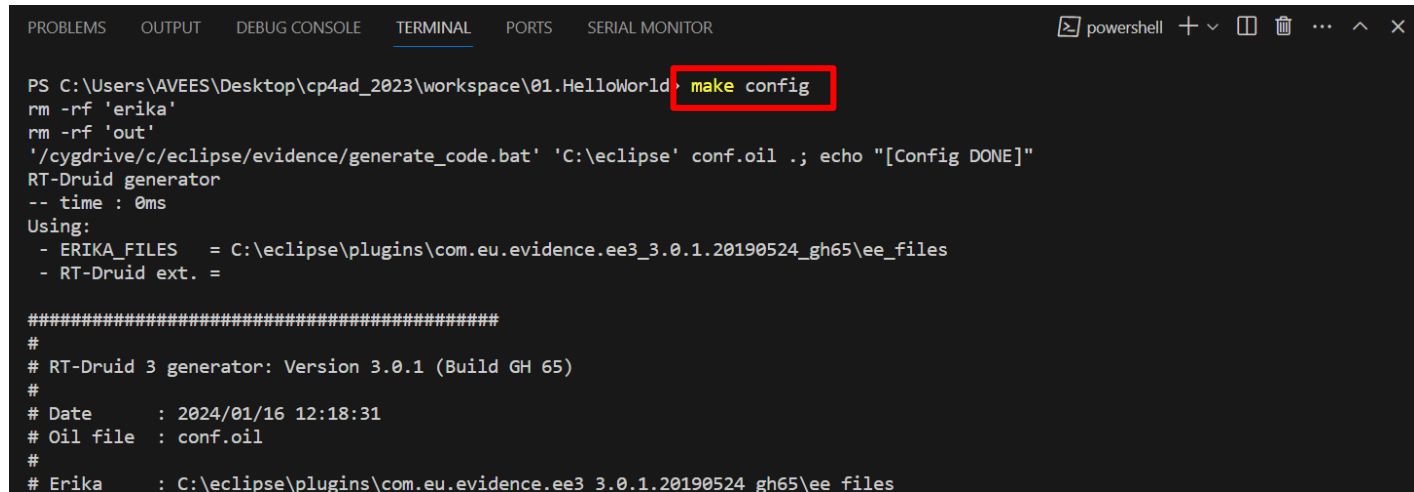
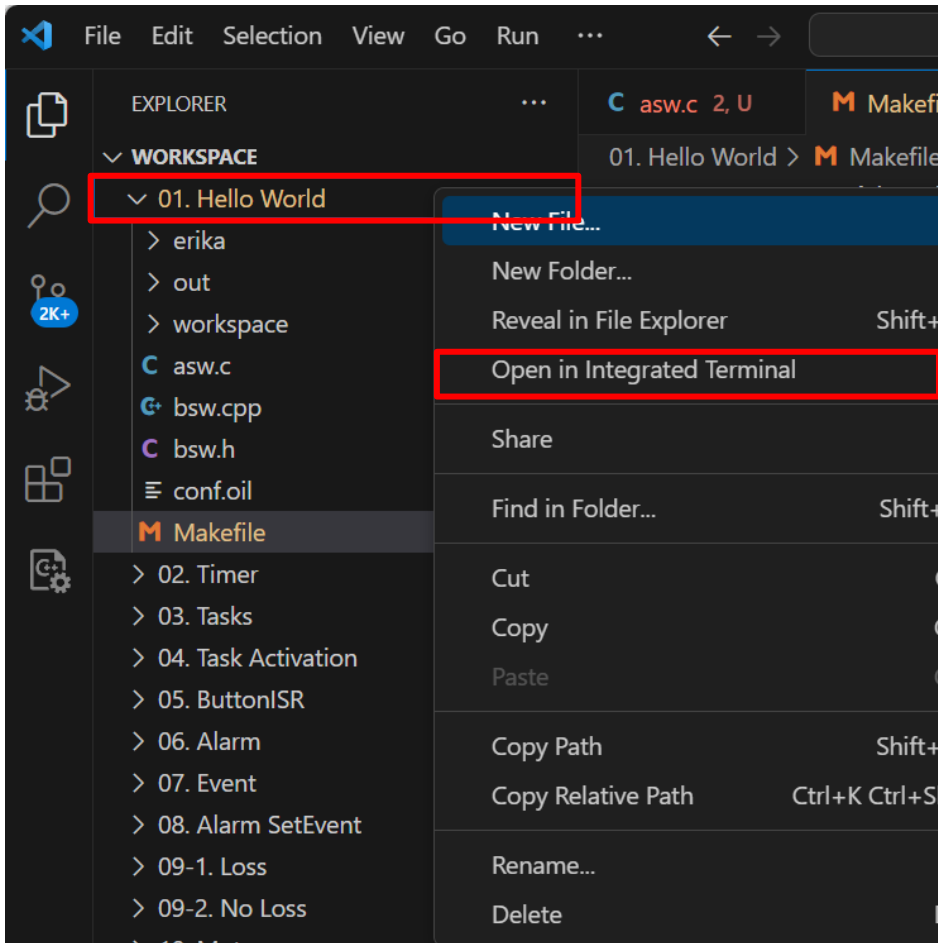
- 01. Hello World/Makefile에서 ARDUINO_PHY_PORT 수정

```
01. Hello World Makefile
1  GENERATOR='/cygdrive/c/eclipse/evidence/generate_code.bat'
2  EE_DIR='C:\eclipse'
3  ARDUINO_DIR="/cygdrive/C/Arduino/hardware/tools/avr/bin"
4  BUILD_DIR='out'
5  ELF_FILE=erika3app.elf
6  ERIKA_DIR='erika'
7  OUT_DIR='out'
8
9  all: build
10
11 build:
12     cd $(BUILD_DIR); sed -i 's/cygpath -ms ""/cygpath -ms "C:\\Arduino/g' makefile; make BINDIR=$(ARDUINO
13
14 touch:
15     touch $(BUILD_DIR)\\erika3app.elf
16
17 config:
18     rm -rf $(ERIKA_DIR)
19     rm -rf $(OUT_DIR)
20     $(GENERATOR) $(EE_DIR) conf.oil .; echo "[Config DONE]"
21
22 clean:
23     cd $(BUILD_DIR); make BINDIR=$(ARDUINO_DIR) clean
24
25 ARDUINO_BAUDRATE    = 115200
26 ARDUINO_MCU         = atmega328p
27
28 ifeq ($(shell uname), Linux)
29 export ARDUINO_SDK_FILES = $(strip )
30 export ARDUINO_PHY_PORT  = /dev/$(ARDUINO_PORT)
31 else # Linux
32 export ARDUINO_SDK_FILES = $(shell cygpath -ms "c:\\Arduino")
33 export ARDUINO_PHY_PORT  = COM8
34 endif # Linux
35
36 PHONY: upload
37 upload:
38     @$(ARDUINO_SDK_FILES)/hardware/tools/avr/bin/avrdude -V \
39     -C$(ARDUINO_SDK_FILES)/hardware/tools/avr/etc/avrdude.conf \
40     -p$(ARDUINO_MCU) -carduino -P$(ARDUINO_PHY_PORT) \
41     -b$(ARDUINO_BAUDRATE) -D -Uflash:w:"$(shell cygpath -wa "$(shell pwd)")\\out\\arduino.hex":i
42
```

Port 번호 맞게 수정

Erika Enterprise Build Process

- 터미널 실행 후 make config 실행



**Build Completed
확인 시 정상 동작**

폴더 우클릭 후 'Open in Integrated Terminal' 실행

Erika Enterprise Build Process

- 터미널에서 make 실행

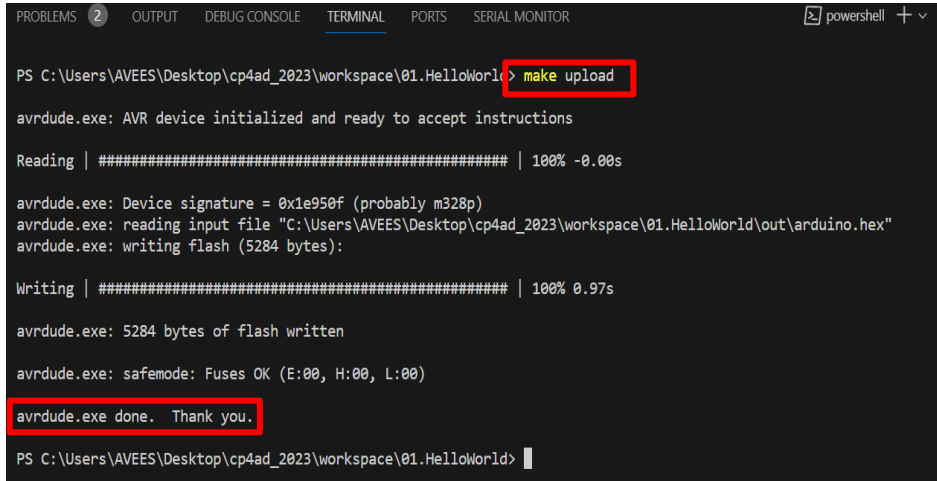
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SERIAL MONITOR
PS C:\Users\AVEES\Desktop\cp4ad_2023\workspace\01.HelloWorld> make
cd 'out'; sed -i 's/cygpath -ms ""/cygpath -ms "C:\\Arduino\\g\\makefile; make BINDIR="/cygdrive/C/Arduino/hardware/tools/avr/bin"
make[1]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/out'
make[2]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/erika'
ERIKA_FILES=/cygdrive/c/eclipse/plugins/COMEUE~4.201/ee_files
make[3]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/erika'
*****
* Pulling erika files from:
*****
* ERIKA_FILES=/cygdrive/c/eclipse/plugins/COMEUE~4.201/ee_files
*****
* Erika Enterprise Pulled successfully! *
*****
make[3]: Leaving directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/erika'
make[2]: Leaving directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/erika'
make[2]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/erika'
ERIKA_FILES=/cygdrive/c/eclipse/plugins/COMEUE~4.201/ee_files
make[3]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/erika'
*****
* Erika Enterprise Libraries build... *
*****
CC hooks.c
CC wiring.c
CC wiring_digital.c
CC wiring_analog.c
```

```
*****
* Erika Enterprise Application built successfully! *
*****
make[2]: Leaving directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/out'
make[1]: Leaving directory '/cygdrive/c/Users/AVEES/Desktop/cp4ad_2023/workspace/01.HelloWorld/out'
PS C:\Users\AVEES\Desktop\cp4ad_2023\workspace\01.HelloWorld>
```

Built successfully
확인 시 정상 동작

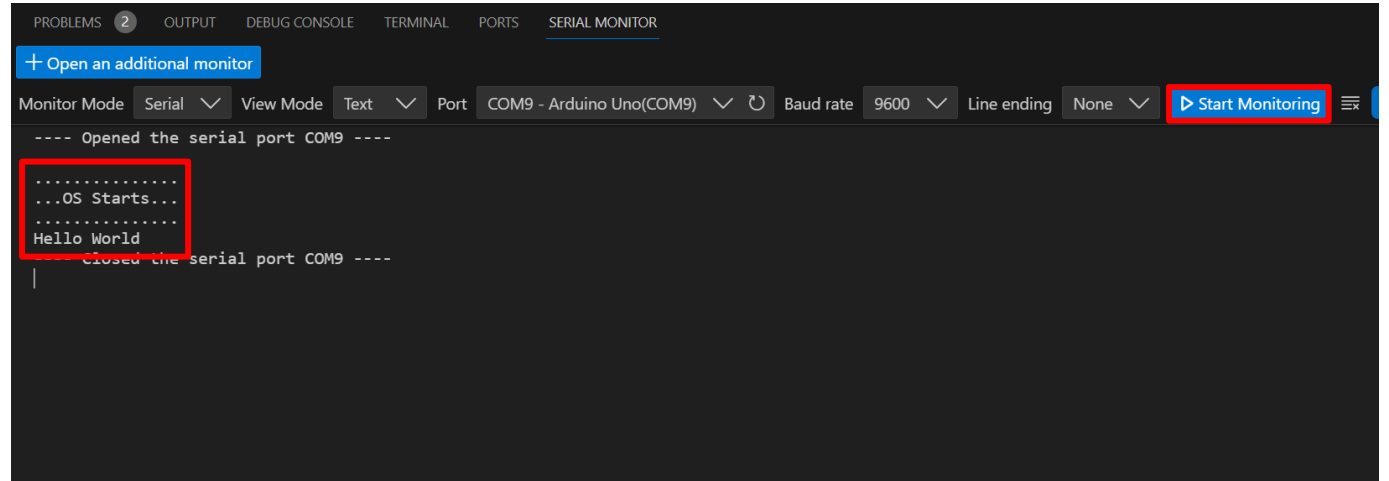
Erika Enterprise Build Process

- make upload 후 Serial Monitor 실행



```
PS C:\Users\AVEES\Desktop\cp4ad_2023\workspace\01.HelloWorld> make upload  
avrdude.exe: AVR device initialized and ready to accept instructions  
  
Reading | ##### | 100% -0.00s  
  
avrdude.exe: Device signature = 0x1e950f (probably m328p)  
avrdude.exe: reading input file "C:\Users\AVEES\Desktop\cp4ad_2023\workspace\01.HelloWorld\out\arduino.hex"  
avrdude.exe: writing flash (5284 bytes):  
  
Writing | ##### | 100% 0.97s  
  
avrdude.exe: 5284 bytes of flash written  
avrdude.exe: safemode: Fuses OK (E:00, H:00, L:00)  
avrdude.exe done. Thank you.  
PS C:\Users\AVEES\Desktop\cp4ad_2023\workspace\01.HelloWorld>
```

Make upload 입력 후 avrdude.exe done 확인

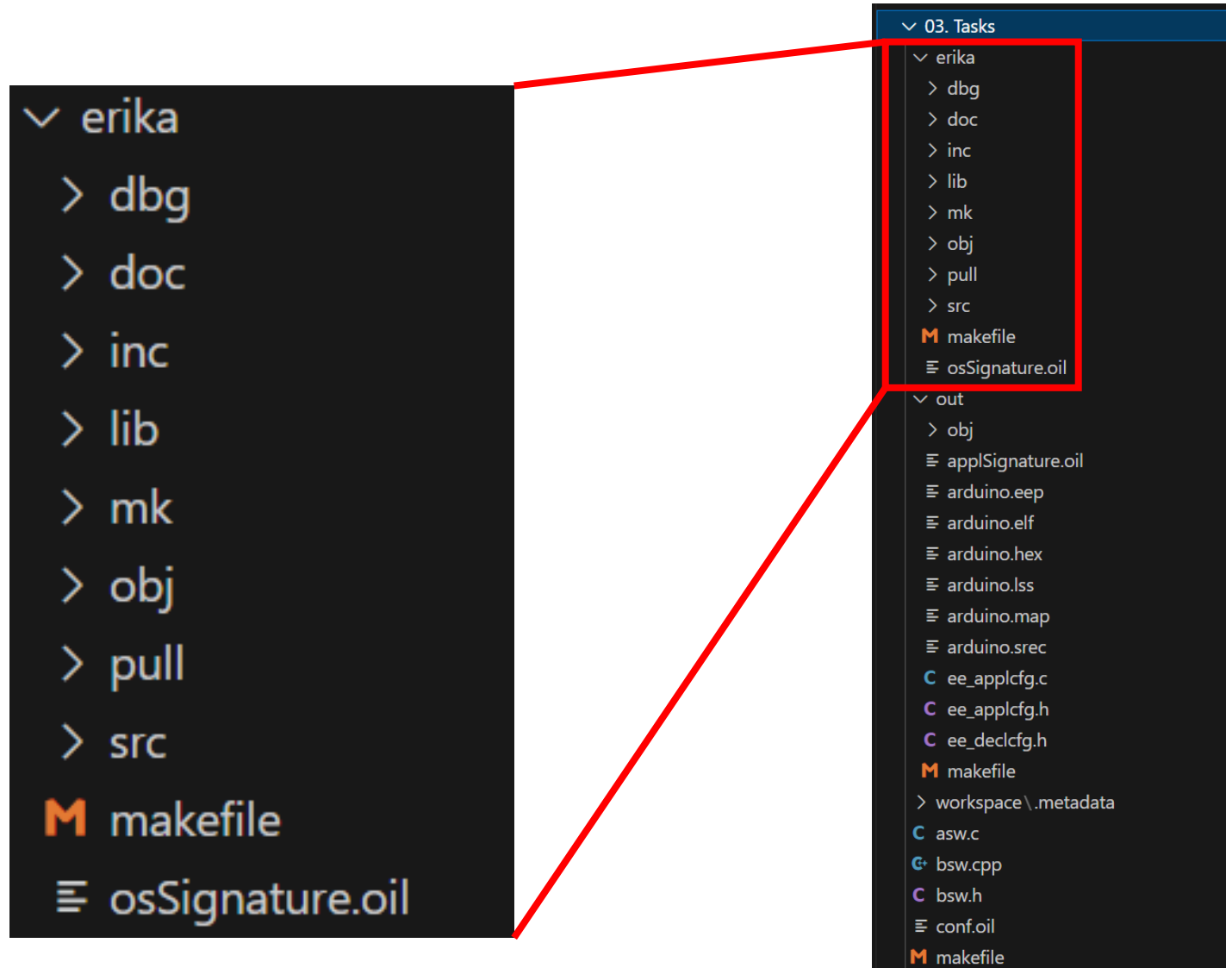


```
+ Open an additional monitor  
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate 9600 Line ending None  
---- Opened the serial port COM9 ----  
.....  
...OS Starts...  
.....  
Hello World  
.....  
---- Closed the serial port COM9 ----  
|
```

Terminal에서 프로그램 동작 확인

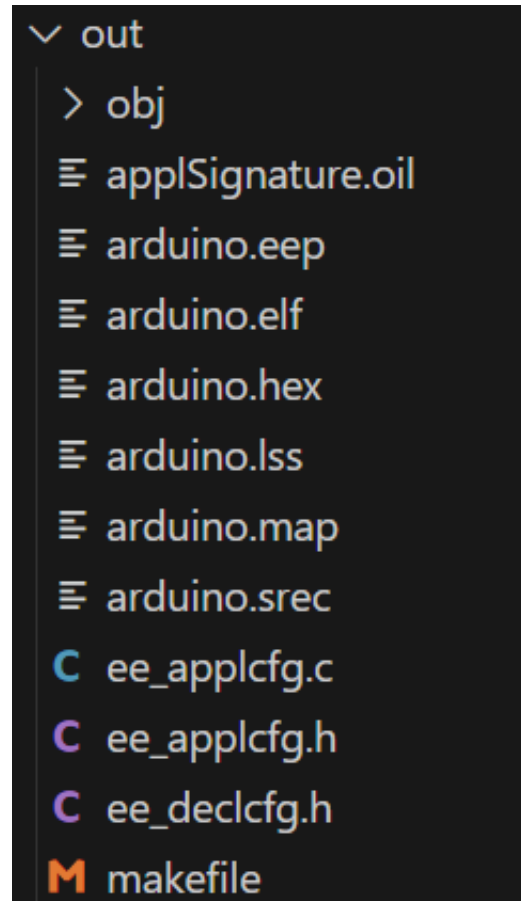
Erika Enterprise Project Structure

- Erika Enterprise 파일
 - 커널 소스
 - 커널 헤더파일

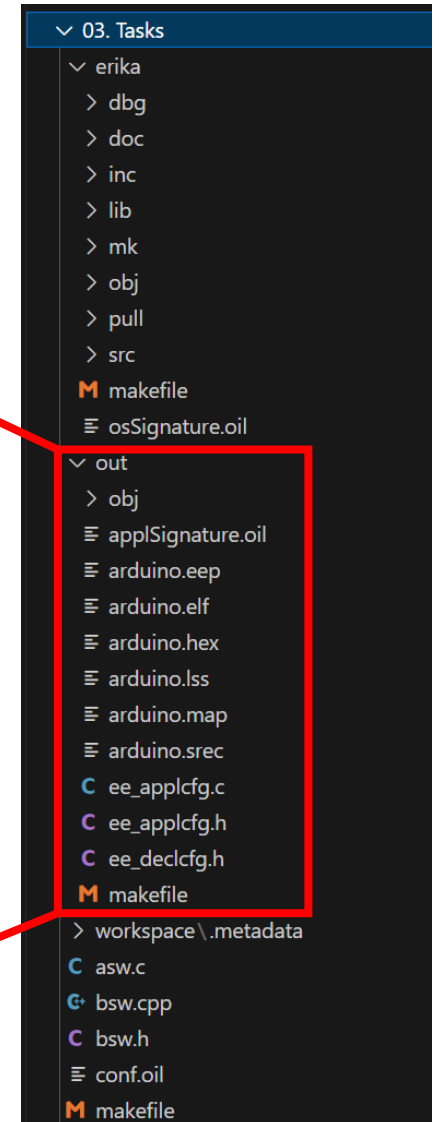


Erika Enterprise Project Structure

- Output 파일
 - OIL에서 자동 생성된 ...cfg.c와 ...cfg.h 파일
 - ELF 파일
 - HEX 파일



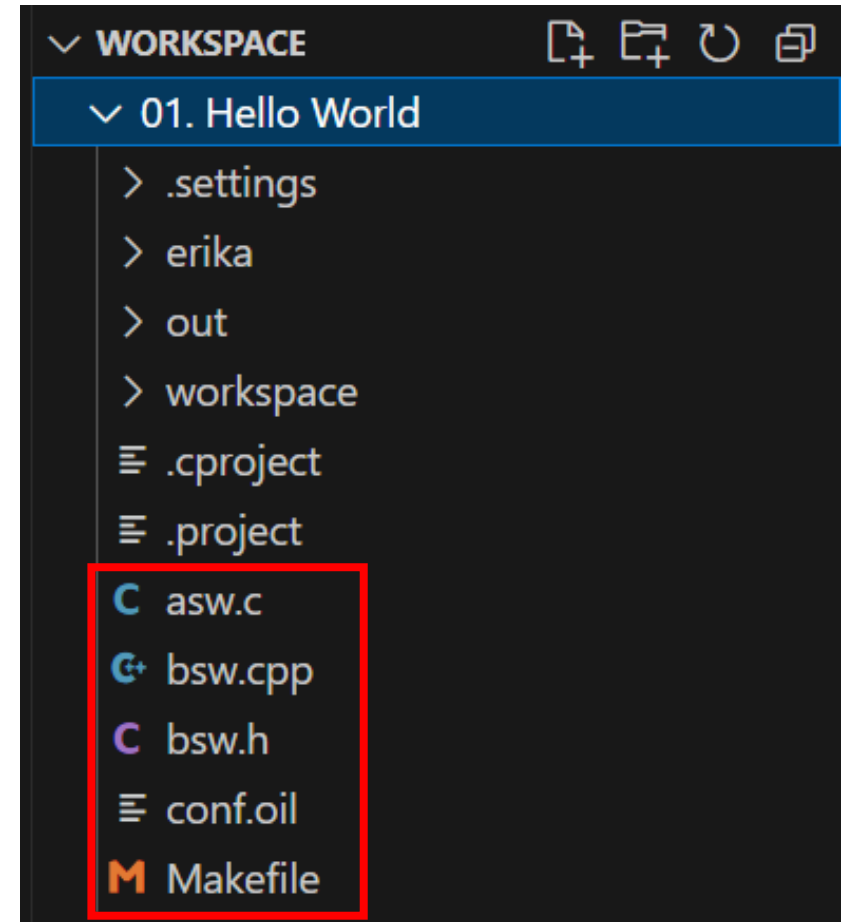
```
out
├── obj
├── applSignature.oil
├── arduino.eep
├── arduino.elf
├── arduino.hex
├── arduino.lss
├── arduino.map
├── arduino.srec
├── ee_applcfg.c
├── ee_applcfg.h
├── ee_declcfg.h
└── makefile
```



```
03. Tasks
├── erika
│   ├── dbg
│   ├── doc
│   ├── inc
│   ├── lib
│   ├── mk
│   ├── obj
│   ├── pull
│   ├── src
│   ├── makefile
│   └── osSignature.oil
├── out
│   ├── obj
│   ├── applSignature.oil
│   ├── arduino.eep
│   ├── arduino.elf
│   ├── arduino.hex
│   ├── arduino.lss
│   ├── arduino.map
│   ├── arduino.srec
│   ├── ee_applcfg.c
│   ├── ee_applcfg.h
│   ├── ee_declcfg.h
│   ├── makefile
│   └── workspace\metadata
├── asw.c
├── bsw.cpp
├── bsw.h
├── conf.oil
└── makefile
```

Erika Enterprise Project Structure

- asw.c
 - 사용자 작성 어플리케이션 코드
- bsw.cpp & bsw.h
 - 사용자 작성 함수
 - main 함수 진입점
- conf.oil
 - OSEK 설정 파일
- Makefile
 - Make 규칙 서술 파일



Erika Enterprise Project Structure

- conf.oil
 - OSEK 설정 파일
 - .oil 확장자, C 구조체 문법 사용 (OSEK Implementation Language)
 - OS object
 - APPDATA object
 - COUNTER object
 - TASK object 등

```
01. Hello World > ≡ conf.oil
1  CPU mySystem {
2      OS myOs {
3          EE_OPT = "DEBUG";
4          USERESSCHEDULER = TRUE;
5          CPU_DATA = AVR8 {
6              MULTI_STACK = FALSE;
7          };
8          MCU_DATA = MEGA {
9              MODEL = MEGA_328p;
10         };
11         LIB = ARDUINO {
12             SDK_BOARD = UNO;
13             VARIANT = CC [1];
14             VERSION = "1.8.16";
15         };
16         STAND_ALONE = TRUE;    // Generate arduino libraries
17     };
18     KERNEL_TYPE = OSEK {
19         CLASS = ECC2;    // Default
20     };
21 };
22
23 APPDATA myApp {
24     APP_SRC = "bsw.cpp";
25     APP_SRC = "asw.c";
26 };
27
28 TASK Task1 {
29     PRIORITY = 1;
30     STACK = SHARED;
31     SCHEDULE = FULL;    // preemptive
32     AUTOSTART = TRUE;
33     ACTIVATION = 1;
34 };
35
```

Erika Enterprise Project Structure

- bsw.cpp & bsw.h
 - Basic SW
 - OS가 시작되는 main() 함수
 - mdelay() ms 단위 delay 함수
 - printfSerial() 콘솔 출력 함수 등

```
01: Hello World > bsw.cpp > main(void)
1  #include "ee.h"
2  #include "Arduino.h"
3  #include "bsw.h"
4
5  #define TIMER1_US 1000000U /* 1 sec */
6
7  #define LEN_BUF 128
8
9  extern "C" {
10
11  void mdelay(unsigned long delay_ms)
12  {
13      unsigned long prev_ms = millis(), current_ms = millis();
14      unsigned long period_ms = 20, cnt = 0;
15      while (cnt < (delay_ms / period_ms)) {
16          current_ms = millis();
17          if (current_ms - prev_ms >= period_ms) {
18              cnt++;
19              prev_ms = millis();
20          }
21      }
22  }
23
24  void printfSerial(const char *fmt, ... )
25  {
26      char buf[LEN_BUF];
27      va_list args;
28      va_start (args, fmt );
29      vsnprintf(buf, LEN_BUF, fmt, args);
30      va_end (args);
31      Serial.print(buf);
32  }
33
34  } /* extern "C" */
35
36  void loop(void)
37  {
38      ;
39  }
40
41  void setup(void)
42  {
43      Serial.begin(9600);
44      OsEE_atmega_startTimer1(TIMER1_US);
45  }
46
47  int main(void)
48  {
49      init();
50
51      #if defined(USBCON)
52      USBDevice.attach();
53      #endif
54
55      setup();
56
57      printfSerial("\n.....\n");
58      printfSerial("...OS Starts...\n");
59      printfSerial(".....\n");
60
61      StartOS(OSDEFAULTAPPMODE); /* never returns */
62
63      return 0;

```


Erika Enterprise Project Structure

- asw.c
 - Application SW
 - 사용자 작성 C 코드 (C++도 가능)
 - BSW 등을 이용하여 Application 레벨에서의 TASK 동작 정의

```
01. Hello World > C asw.c > TASK(Task1)
1  #include "bsw.h"
2
3  TASK(Task1)
4  {
5      // Write the code below.
6      printfSerial("Hello World\n");
7
8      TerminateTask();
9  }
10
```

Erika Enterprise Project Structure

- Makefile
 - Make 규칙 서술 파일
 - make config
out, erika 폴더 삭제 후
conf.oil 파일을 기반으로 코드 생성
 - make (build)
프로젝트 빌드
 - make upload
빌드 결과 생성된 .hex파일을
Arduino 보드에 업로드

```
01. Hello World > M Makefile
1  GENERATOR='/cygdrive/c/eclipse/evidence/generate_code.bat'
2  EE_DIR='C:\eclipse'
3  ARDUINO_DIR="/cygdrive/C/Arduino/hardware/tools/avr/bin"
4  BUILD_DIR='out'
5  ELF_FILE=erika3app.elf
6  ERIKA_DIR='erika'
7  OUT_DIR='out'
8
9  all: build
10
11 build:
12     cd $(BUILD_DIR); sed -i 's/cygpath -ms ""/cygpath -ms "C:\\Arduino/g' makefile; make BINDIR=$(ARDUINO_DIR)
13
14 touch:
15     touch $(BUILD_DIR)\\erika3app.elf
16
17 config:
18     rm -rf $(ERIKA_DIR)
19     rm -rf $(OUT_DIR)
20     $(GENERATOR) $(EE_DIR) conf.oil .; echo "[Config DONE]"
21
22 clean:
23     cd $(BUILD_DIR); make BINDIR=$(ARDUINO_DIR) clean
24
25 ARDUINO_BAUDRATE    = 115200
26 ARDUINO_MCU         = atmega328p
27
28 ifeq ($(shell uname), Linux)
29 export ARDUINO_SDK_FILES = $(strip )
30 export ARDUINO_PHY_PORT  = /dev/$(ARDUINO_PORT)
31 else # Linux
32 export ARDUINO_SDK_FILES = $(shell cygpath -ms "c:\\Arduino")
33 export ARDUINO_PHY_PORT  = COM8
34 endif # Linux
35
36 PHONY: upload
37 upload:
38     @$ (ARDUINO_SDK_FILES)/hardware/tools/avr/bin/avrdude -V \
39     -C$(ARDUINO_SDK_FILES)/hardware/tools/avr/etc/avrdude.conf \
40     -p$(ARDUINO_MCU) -carduino -P$(ARDUINO_PHY_PORT) \
41     -b$(ARDUINO_BAUDRATE) -D -Uflash:w:"$(shell cygpath -wa "$(shell pwd)")\\out\\arduino.hex":i
42
```

02. Timer

- asw.c 파일에 ISR2로 빈 TimerISR 콜백함수 추가
- OIL 파일에 TimerISR 추가
 - Category 2
 - Timer 1 Compare Match A

asw.c

```
ISR2(TimerISR)
{
    ;
}
```

conf.oil

```
ISR TimerISR {
    CATEGORY = 2;
    SOURCE = "TIMER1_COMPA";
};
```

02. Timer

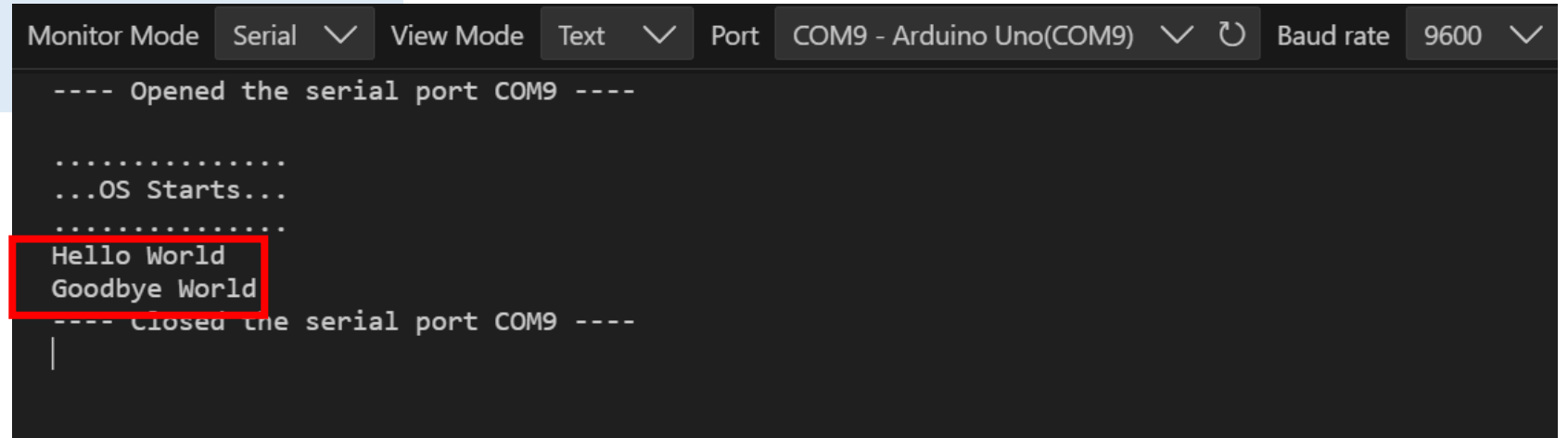
- mdelay 함수 이용하여 3초의 Task1 실행시간 표현

asw.c

```
TASK(Task1)
{
    printfSerial("Hello World\n");

    mdelay(3000);

    printfSerial("Goodbye World\n");
    TerminateTask();
}
```

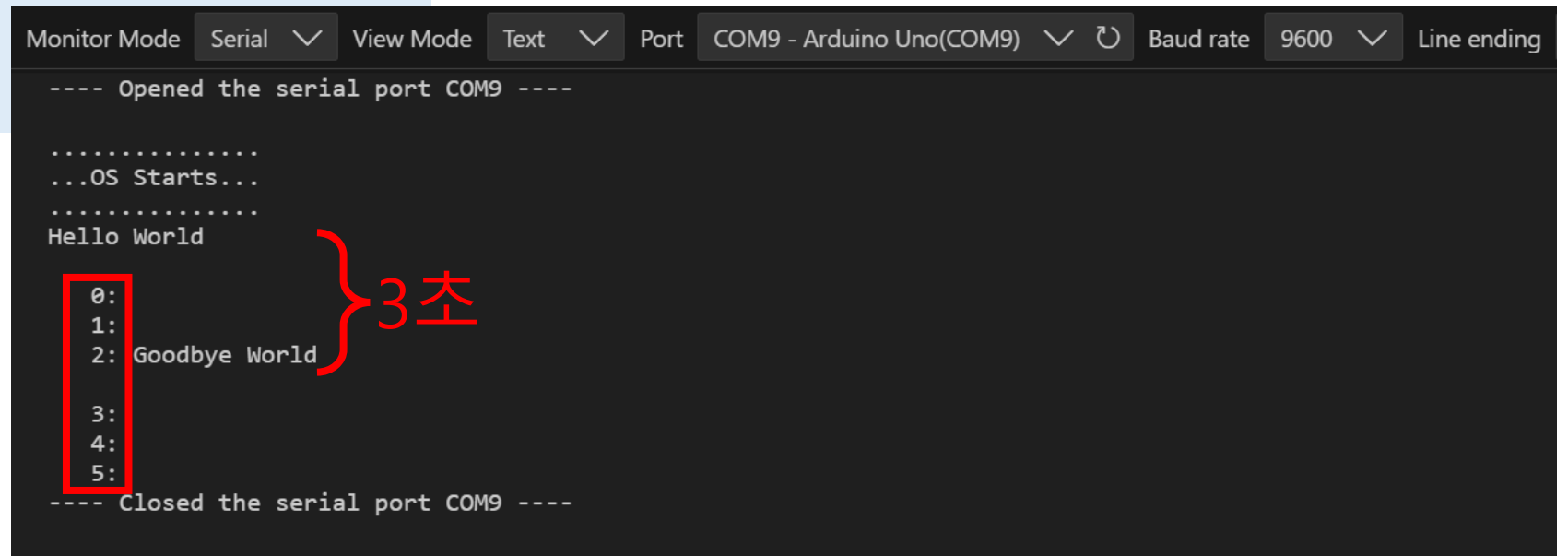


```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate 9600
---- Opened the serial port COM9 ----
.....
...OS Starts...
.....
Hello World
Goodbye World
---- Closed the serial port COM9 ----
|
```

02. Timer

- TimerISR 이용 초단위 시간 출력
asw.c

```
ISR2(TimerISR)
{
    static long c = 0;
    printfSerial("\n%4ld: ", c++);
}
```



```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate 9600 Line ending
---- Opened the serial port COM9 ----
.....
...OS Starts...
.....
Hello World
0:
1:
2: Goodbye World
3:
4:
5:
---- Closed the serial port COM9 ----
```

03. Tasks

asw.c

```
TASK(Task1)
{
    printfSerial("Task1 Begins...");
    mdelay(3000);
    printfSerial("Task1 Finishes...");

    TerminateTask();
}

TASK(Task2)
{
    printfSerial("Task2 Begins...");
    mdelay(3000);
    printfSerial("Task2 Finishes...");

    TerminateTask();
}
```

- 높은 우선순위의 Task2 추가

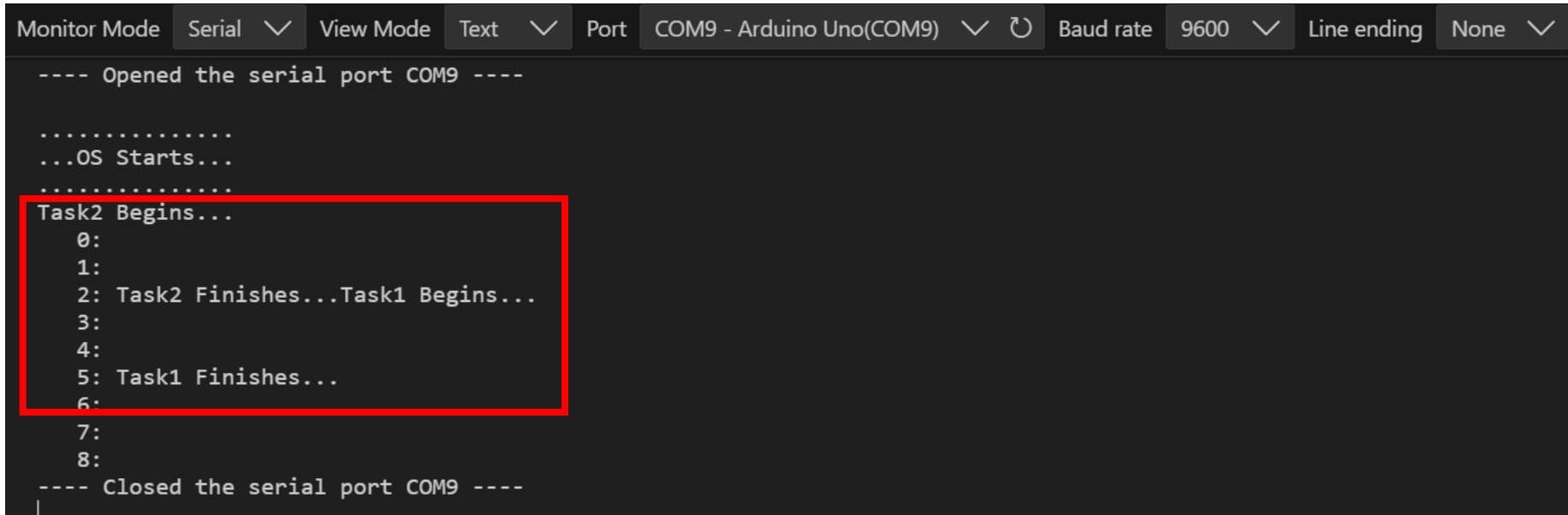
conf.oil

```
TASK Task2
{
    PRIORITY = 2;
    STACK = SHARED;
    SCHEDULE = FULL; // preemptive
    AUTOSTART = TRUE;
    ACTIVATION = 1;
};
```

클수록 높은
우선순위

03. Tasks

- Task2가 먼저 시작, Task1 대기
- Task2 종료 후 Task1 시작



The screenshot shows the serial monitor interface with the following text:

```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate 9600 Line ending None  
---- Opened the serial port COM9 ----  
.....  
...OS Starts...  
.....  
Task2 Begins...  
0:  
1:  
2: Task2 Finishes...Task1 Begins...  
3:  
4:  
5: Task1 Finishes...  
6:  
7:  
8:  
---- Closed the serial port COM9 ----
```

A red rectangular box highlights the lines from "Task2 Begins..." to "6:", indicating the period where Task2 is running and Task1 is waiting.

- 우선순위를 바꾼다면?

04. Task Activation

asw.c

```
ISR2(TimerISR)
{
    static long c = -4;
    if (c == 0)
        ActivateTask(Task1);
    printfSerial("\n%4ld: ", c++);
}
```

```
TASK(Task1)
{
    printfSerial("Task1 Begins...");
    mdelay(3000);
    ActivateTask(Task2);
    mdelay(3000);
    printfSerial("Task1 Finishes...");
    TerminateTask();
}
```

Task 활성화

```
TASK(Task2)
{
    printfSerial("Task2 Begins...");
    mdelay(3000);
    printfSerial("Task2 Finishes...");
    TerminateTask();
}
```

conf.oil

```
TASK Task1 {
    ...
    AUTOSTART = FALSE;
    ...
};

TASK Task2 {
    ...
    AUTOSTART = FALSE;
    ...
};
```

Task의
자동실행 Off

04. Task Activation

- -4부터 카운트다운, 0에서 TimerISR에 의해 ActivateTask(Task1) 실행
- 3초 후 ActivateTask(Task2) 실행, Task2의 Task1 선점 확인



```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate 9600 Line ending
---- Opened the serial port COM9 ----

.....
...OS Starts...
.....

-4:
-3:
-2:
-1:
0: Task1 Begins...
1:
2:
3: Task2 Begins...
4:
5:
6: Task2 Finishes...
7:
8:
9: Task1 Finishes...
10:
11:
12:
---- Closed the serial port COM9 ----
```

- 우선순위가 바뀌면?

05. ButtonISR

- bsw.cpp 수정: 버튼 인터럽트 관련 코드
- OIL 수정: ButtonISR 추가

bsw.cpp

```
...
#include <avr/io.h>
#include <avr/interrupt.h>
...
#define _BV(bit) (1 << (bit))
int main(void)
{
    PORTC = _BV(PC0);
    PCICR = _BV(PCIE1);
    PCMSK1 = _BV(PCINT8);
    EIFR = 0xff;
    sei();

    init();
}
```

conf.oil

```
ISR ButtonISR {
    CATEGORY = 2;
    SOURCE = "PCINT1";
};
```

05. ButtonISR

asw.c

```
ISR2(TimerISR)
{
    static long c = -4;
if (c == 0)
    ActivateTask(Task1);
    printfSerial("\n%4ld: ", c++);
}
```

asw.c

```
ISR2(ButtonISR)
{
    int a0;
    DisableAllInterrupts();
    if ((PINC & 0x01) != 0) return;
    printfSerial("<BUTTON ISR>");
    a0 = analogRead(A0); // read ADC value
    if (a0 < 50) { // UP
        ActivateTask(Task1);
    } else if (a0 < 200) { // DOWN
        ActivateTask(Task2);
    } else if (a0 < 380) { // LEFT
        ;
    } else if (a0 < 520) { // RIGHT
        ;
    }
    EnableAllInterrupts();
}
```

05. ButtonISR

- 중복 Activation의 경우?
 - ACTIVATION = 1; 수정 필요
- ISR에서 mdelay 실행하면?

```
ISR2(ButtonISR)
{
    int a0;
    //DisableAllInterrupts();
    if ((PINC & 0x01) != 0) return;
    printfSerial("<BUTTON ISR>");
    a0 = analogRead(A0); // read ADC value
    if (a0 < 50) { // UP
        ActivateTask(Task1);
    }
    else if (a0 < 200) { // DOWN
        ActivateTask(Task2);
    }
    else if (a0 < 380) { // LEFT
        ;
    }
    else if (a0 < 520) { // RIGHT
        ;
    }
    mdelay(3000);
    printfSerial("<----->")
    //EnableAllInterrupts();
}
```

```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate
---- Opened the serial port COM9 ----

.....
...OS Starts...
.....

-4:
-3:
-2: <BUTTON ISR>Task1 Begins...
-1:
0: Task2 Begins...
1: Task2 Begins...
2:
3:
4: Task2 Finishes...
5:
6: <BUTTON ISR>Task2 Begins...
7:
8:
9: Task2 Finishes...
10: Task1 Finishes...
11:
12:
---- Closed the serial port COM9 ----
```

Questions

